In this paper, every Ring is Commutative Ring with $1 \neq 0$.

Thm. Let $A_1, \cdots, A_k$ be Ideals in R. Define a map

$$\varphi : R \to R/A_1 \times \cdots \times R/A_k : r \mapsto (r+A_1, \cdots, r+A_k)$$

is a Ring Homomorphism with $\ker \varphi = A_1 \cap A_2 \cap \cdots \cap A_k$.
If for each $1 \leq i < j \leq k$, $A_i + A_j = R$ (comaximal), then φ is onto and

$$\ker \varphi = A_1 \cap \cdots \cap A_k = A_1 A_2 \cdots A_k.$$

Therefore, $R/(A_1 \cdots A_k) = R/(A_1 \cap \cdots \cap A_k) \cong R/A_1 \times \cdots \times R/A_k$.
Proof. First, prove this for $k=2$. The map

$$\varphi : R \to R/A_1 \times R/A_2 : r \mapsto (r+A_1, r+A_2)$$

is just the natural projection for R/A_1, R/A_2 elementwise, so ring-Homomorphism.
Moreover,

$$\varphi(r) = 0 \iff r + A_1 = A_1 \text{ and } r + A_2 = A_2$$
$$\iff r \in A_1 \text{ and } r \in A_2$$

so $\ker \varphi = A_1 \cap A_2$. Now, suppose that $A_1 + A_2 = R$. Since $1 \in R = A_1 + A_2$,

$$x + y = 1 \text{ for some } x \in A_1 \text{ and } y \in A_2.$$

Therefore, $\varphi(x) = \varphi(1-y) = (x + A_1, 1-y+A_2) = (0+A_1, 1+A_2)$.
And similarly $\varphi(y) = (1, 0)$. Now, given $(r_1 + A_1, r_2 + A_2) \in R/A_1 \times R/A_2$,

$$\varphi(r_2 x + r_1 y) = \varphi(r_2)\varphi(x) + \varphi(r_1)\varphi(y)$$
$$= \varphi(r_2)(\bar{0}, \bar{1}) + \varphi(r_1)(\bar{1}, \bar{0})$$
$$= (\bar{0}, \bar{r_2}) + (\bar{r_1}, \bar{0}) = (r_1 + A_1, r_2 + A_2).$$

Thus φ is onto. And $A_1 A_2 \subseteq A_1 \cap A_2$, because: for $a_i \in A_1$, $b_i \in A_2$,

$$a_1 b_1 + \cdots + a_n b_n \in A_1 \text{ and } A_2 \text{ since } A_1, A_2 \text{ are Ideal.}$$

Meanwhile, if $A_1 + A_2 = R$, for any $c \in A_1 + A_2$,

$$c = c \cdot 1 = c(x+y) = cx + cy \in A_1 + A_2.$$

So proved.

Thm. Let R be a Ring with identity $1 \neq 0$.
Let $e \in R$ be an idempotent in R such that $er = re$ for all $r \in R$.
Then, Re and $R(1-e)$ are (two-sided) ideal of R and that

$$R \cong Re \times R \cdot (1-e)$$

Moreover, e and $1-e$ are identity for Re and $R(1-e)$, respectively.

Proof. Notice e commutes with respect to all elements in R, $Re = eR$.
and given $r_1, r_2 \in R$, $r_1 e - r_2 e = (r_1 - r_2)e \in Re$, so Re is subgroup
and given $r \in R$ and $r'e \in Re$, $rr'e \in Re$, so Re is left-sided ideal.
Since $eR = Re$, similarly it is right-sided ideal.
Moreover, $1-e$ also commutes in R, so $R(1-e)$ is ideal, similarly Re.
Given $r \in R$, $r = re + r - re$, so $R = Re + R(1-e)$, comaximal.
By the Chinese Remainder Theorem, $Re \cap R(1-e) = Re R(1-e)$.
observe that: given $r_1, r_2 \in R$,

$$r_1 e \cdot (r_2 - r_2 e) = r_1 r_2 e - r_1 r_2 e^2 = r_1 r_2 (e - e^2) = r_1 r_2 0 = 0,$$

Therefore, $Re \cdot R(1-e) = \{0\}$, Meanwhile,

$$\varphi : R \to Re ; r \mapsto re$$

is onto Ring-Homomorphism, because: Surjective is trivial and

$$\varphi(r_1 + r_2) = (r_1 + r_2)e = r_1 e + r_2 e = \varphi(r_1) + \varphi(r_2) \text{ and}$$

$$\varphi(r_1 r_2) = r_1 r_2 e = r_1 r_2 e^2 = r_1 e r_2 e = \varphi(r_1)\varphi(r_2)$$

And $\varphi(r) = re = 0 \iff r \in R(1-e)$, because
$re = 0 \Rightarrow r = (r-e) - (r-e) \cdot e \in R(1-e)$
$r \in R(1-e) \Rightarrow r = r' - r'e$ for some $r' \in R \Rightarrow re = r'e - r'e^2 = 0$. So $\ker \varphi = R(1-e)$.
So, $Re \cong R/R(1-e)$ and similarly $R(1-e) \cong R/Re$.
Consequently, the Chinese Remainder theorem gives

$$R \cong R/\{0\} \cong (R/R(1-e)) \times (R/Re) \cong Re \times R(1-e)$$

Further given $re \in Re$, $ree = ere = re^2 = re$, so e is identity in Re,
similarly $1-e$ is identity of $R(1-e)$. ☒

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Then, Re and $R(1-e)$ are (two-sided) ideal of R and that

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Moreover, e and $1-e$ are identity for Re and $R(1-e)$, respectively.

Proof. Notice e commutes with respect to all elements in R, $Re = eR$.
and given $r_1, r_2 \in R$, $r_1e - r_2e = (r_1 - r_2)e \in Re$, so Re is subgroup
and given $r \in R$ and $r'e \in Re$, $rr'e \in Re$, so Re is left-sided ideal.
Since $eR = Re$, similarly it is right-sided ideal.
And, since $(1-e)^2 = 1 - e + e - e^2 = 1-e$, so $1-e$ is also idempotent and
$1-e$ also commute with all elements in R, so $R(1-e)$ is also ideal.
And, given $re \in Re$, $re \cdot e = re = e \cdot re$, so e is identity of Re,
and given $r(1-e) \in R(1-e)$, $r(1-e) \cdot (1-e) = r(1-e+e-e^2) = r(1-e) = (1-e)r(1-e)$,
So $1-e$ is an identity of $R(1-e)$.

Define a map $\varphi : R \to Re \times R(1-e) : r \mapsto (re, r-re)$. Then,

$$\varphi(r_1+r_2) = ((r_1+r_2)e, (r_1+r_2) - (r_1+r_2)e)$$
$$= (r_1e, r_1 - r_1e) + (r_2e, r_2 - r_2e) = \varphi(r_1) + \varphi(r_2)$$
$$\varphi(r_1r_2) = (r_1r_2e, r_1r_2 - r_1r_2e)$$
$$= (r_1er_2e, (r_1 - r_1e)(r_2 - r_2e)) = \varphi(r_1)\varphi(r_2)$$

Therefore, φ is Ring-Homomorphism and onto, because:
given $(r_1e, r_2(1-e)) \in Re \times R(1-e)$, let $r = r_1e + r_2(1-e)$. Then,
$\varphi(r) = (r_1e^2 + r_2(1-e)\cdot e, r_1e(1-e) + r_2(1-e)^2) = (r_1e, r_2(1-e))$.

Moreover, $\ker \varphi = \{r \in R \mid \varphi(r) = (0,0)\}$, and $re = 0$ and $r(1-e) = 0$ implies
$re + r(1-e) = r = 0$, so $\ker \varphi = \{0\}$. So the first isomorphism theorem
gives that $R / \ker \varphi \cong R \cong Re \times R(1-e)$ $\boxtimes$

Prop. If R and S are non-zero rings, then $R \times S$ is never a field.
Proof. Define a map

$$\varphi : R \times S \to R ; (r, s) \mapsto r.$$

Then, $\varphi((r_1, s_1) \cdot (r_2, s_2)) = \varphi(r_1 r_2, s_1 s_2) = r_1 r_2 = \varphi(r_1, s_1) \varphi(r_2, s_2)$,
and $\varphi((r_1, s_1) + (r_2, s_2)) = \varphi((r_1 + r_2, s_1 + s_2)) = r_1 + r_2 = \varphi(r_1, s_1) + \varphi(r_2, s_2)$
So, φ is Ring Homomorphism and trivially onto,
More over, $\operatorname{Ker} \varphi = \{(r, s) \in R \times S \mid r = 0\} = \{0\} \times S$.
So, $R \times S$ has non-trivial proper ideal, So it is never a field. $\blacksquare$